Kuboid va kubning bir yog'i yuzini topish

2-test (qiyinroq) · 20 ta savol

Ism-familiya: _____ Sinf: _____ Sana: _____

Har savolda bitta to'g'ri javobni belgilang.

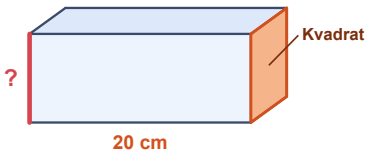
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1. **Kuboidning hajmi 1920 cm³. Ust yog'ining yuzi 160 cm², old yog'ining yuzi 96 cm². Yon yog'ining yuzi qancha?**

A) 256 cm² **B)** 120 cm² **C)** 240 cm² **D)** 64 cm²

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2. **Rasmdagi kuboidning hajmi 1280 cm³. Bo'yalgan yon yog'i kvadrat. Kvadratning tomoni qancha?**



A) 64 cm **B)** 8 cm

C) 4 cm **D)** 32 cm

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3. **Kubning bir yog'i yuzi 676 cm². Kubning hajmi qancha? (Kalkulyatordan foydalanish mumkin.)**

A) 4056 cm³ **B)** 17 576 cm³

C) 456 976 cm³ **D)** 2028 cm³

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4. **A kubning hajmi 512 000 cm³. B kubning qirrasi A kubning qirrasidan 4 marta qisqa. B kubning hajmi qancha?**

A) 128 000 cm³ **B)** 8000 cm³

C) 2000 cm³ **D)** 32 000 cm³

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5. **Kuboid shaklidagi idishning balandligi 60 cm, asosi kvadrat. Idishning $\frac{1}{3}$ qismi to'lishi uchun 18 l suv ketdi. Asosidagi kvadratning tomoni qancha?**

A) 30 cm **B)** ≈ 17,3 cm **C)** 900 cm **D)** 10 cm

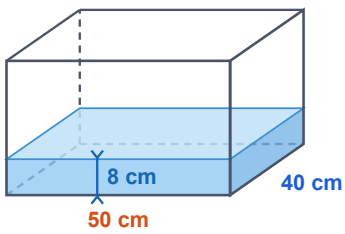
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6. **Akvariumning bo'yi 60 cm, eni 35 cm, balandligi 40 cm. Unda 31,5 l suv bor. Akvarium to'lishi uchun yana necha litr suv kerak?**

A) 52,5 l **B)** 84 l **C)** 31,5 l **D)** 25 l

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7. **A idishning bo'yi 30 cm, eni 20 cm, balandligi 25 cm, u suvga to'la. Uning hamma suvi rasmdagi B idishga quyildi. B dagi suv sathi qancha bo'ldi?**



A) 33 cm **B)** 23 cm

C) 7,5 cm **D)** 15,5 cm

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8. **Akvariumning bo'yi 40 cm, eni 30 cm, balandligi 30 cm, uning $\frac{2}{3}$ qismida suv bor. Akvariumni yuvish uchun hamma suv tubi 60 cm ga 40 cm bo'lgan bo'sh bakka quyib qo'yildi. Bakda suv qanday balandlikda turadi?**

A) 30 cm **B)** 10 cm **C)** 20 cm **D)** 15 cm

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9. **Bo'sh idishning bo'yi 80 cm, eni 50 cm, balandligi 60 cm. Unga har minutda 20 l suv quyiladi. Idish yarmigacha necha minutda to'ladi?**

A) 60 min **B)** 12 min **C)** 3 min **D)** 6 min

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10. **Bo'sh idishning tubi 120 cm ga 50 cm. Jo'mrakdan har minutda 30 l suv tushib, idish 20 minutda to'ldi. Idishning balandligi qancha?**

A) 10 cm **B)** 100 cm **C)** 1000 cm **D)** 50 cm

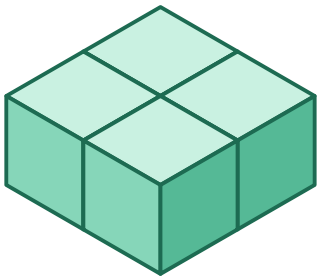
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11. **Hovuzning tubi 10 m ga 4 m, undagi suv sathi 1,5 m. Hovuzdan 12 m³ suv chiqarildi. Endi suv sathi qanday balandlikda?**

A) 0,3 m **B)** 1,2 m **C)** 1,8 m **D)** 1,38 m

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12. **Rasmdagi jism har birining hajmi 8 cm³ bo'lgan 4 ta kubdan yasalgan. Jismning hamma tomoni (pasti ham) bo'yaldi. Bo'yalgan yuz qancha?**



A) 32 cm² **B)** 64 cm²

C) 16 cm² **D)** 96 cm²

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13. **Qirrasi 4 cm bo'lgan kub shaklidagi qand bo'laklari ichki o'lchamlari 30 cm, 20 cm va 10 cm bo'lgan qutiga teriladi. Qutiga ko'pi bilan nechta qand bo'lagi sig'adi?**

A) 60 **B)** 75 **C)** 70 **D)** 93

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14. **Ikki g'ishtning hajmi bir xil. Birinchisining o'lchamlari 25 cm, 8 cm va 5 cm. Ikkinchisining uzunligi 10 cm, ikki uchidagi yoqlari kvadrat. Kvadratning tomoni qancha?**

A) 25 cm **B)** 10 cm **C)** 50 cm **D)** 100 cm

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15. **Bo'yi 36 cm bo'lgan kuboid 3 ta bir xil kubga arralandi. Kuboidning hajmi qancha edi?**

A) 5184 cm³ **B)** 46 656 cm³

C) 432 cm³ **D)** 1728 cm³

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16. **Devorga uzunligi 12 cm bo'lgan tosh blok va uning uchiga hajmi 125 cm³ bo'lgan kub shaklidagi tosh terildi. Blokning uchidagi yog'i kub yog'i bilan aynan bir xil kvadrat, ikkalasi birga bitta kuboid shaklida bo'ldi. Shu kuboidning hajmi qancha?**

A) 300 cm³ **B)** 2125 cm³ **C)** 425 cm³ **D)** 137 cm³

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17. **Ikki idish pastdan quvur bilan ulangan, quvur yopiq. A idish tubining yuzi 300 cm², unda 40 cm balandlikda suv bor. B idish tubining yuzi 500 cm², u bo'sh. Quvur ochilgach, suv B ga o'tib, ikkala idishdagi sath tenglashdi. Sath qanday balandlikda?**

A) 25 cm **B)** 20 cm **C)** 24 cm **D)** 15 cm

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18. **Idishning $\frac{3}{5}$ qismida 36 l suv bor. Idish to'lishi uchun yana necha litr suv kerak?**

A) 21,6 l **B)** 24 l **C)** 14,4 l **D)** 60 l

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19. Idishning tubi 45 cm ga 40 cm. Unga suv quyilib, sath 12 cm dan 20 cm ga ko'tarildi. Necha litr suv quyildi?

A) 14,4 l **B)** 21,6 l **C)** 36 l **D)** 144 l

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20. Kubning hajmi 32 768 cm³. Uning bir yog'i yuzi qancha? (Kalkulyatordan foydalanish mumkin.)

A) 3072 cm² **B)** 2048 cm² **C)** 32 cm² **D)** 1024 cm²

Javoblar

Kuboid va kubning bir yog'i yuzini topish · 2-test (qiyinroq)

№	Javob	Qiymati
1	**C**	240 cm²
2	**B**	8 cm
3	**B**	17 576 cm³
4	**B**	8000 cm³
5	**A**	30 cm
6	**A**	52,5 l
7	**D**	15,5 cm
8	**B**	10 cm
9	**D**	6 min
10	**B**	100 cm

№	Javob	Qiymati
11	**B**	1,2 m
12	**B**	64 cm²
13	**C**	70
14	**B**	10 cm
15	**A**	5184 cm³
16	**C**	425 cm³
17	**D**	15 cm
18	**B**	24 l
19	**A**	14,4 l
20	**D**	1024 cm²

Qisqa yechimlar

1. Balandligi: 1920 : 160 = 12 cm. Eni: 1920 : 96 = 20 cm.
Yon yoq eni va balandligidan iborat: 20 · 12 = **240 cm²**.
(256 — ikki yuz qo'shilgan.)

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2. Kvadrat yuzi: 1280 : 20 = 64 cm².
Tomoni: 8 · 8 = 64, demak **8 cm**.
(32 — 64 ning yarmi.)

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3. Qirrasi: 26 · 26 = 676, demak 26 cm.
Hajmi: 26 · 26 · 26 = **17 576 cm³**.
(4056 — 676 · 6, 456 976 — 676 · 676.)

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4. A qirrasi: 80 · 80 · 80 = 512 000, demak 80 cm.
B qirrasi: 80 : 4 = 20 cm. Hajmi: 20 · 20 · 20 = **8000 cm³**.
(128 000 — hajm 4 ga bo'lingan; qirra 4 marta qisqarsa, hajm 4 · 4 · 4 = 64 marta kamayadi.)

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5. Suv sathi: 60 : 3 = 20 cm. 18 l = 18 000 cm³.
Asosi: 18 000 : 20 = 900 cm². Tomoni: 30 · 30 = 900, demak **30 cm**.
(≈ 17,3 — 18 000 butun balandlik 60 ga bo'lingan.)

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6. Tubi: 60 · 35 = 2100 cm². Sath: 31 500 : 2100 = 15 cm.
Bo'sh qism: 40 − 15 = 25 cm, hajmi 2100 · 25 = 52 500 cm³ = **52,5 l**.
(84 l — butun akvarium.)

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7. A dagi suv: 30 · 20 · 25 = 15 000 cm³. B tubi: 50 · 40 = 2000 cm².
Sath 15 000 : 2000 = 7,5 cm ga ko'tariladi: 8 + 7,5 = **15,5 cm**.
(7,5 — B dagi oldingi 8 cm qo'shilmagan.)

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8. Akvariumdagi sath: 30 · $\frac{2}{3}$ = 20 cm. Suv: 40 · 30 · 20 = 24 000 cm³.
Bak tubi: 60 · 40 = 2400 cm². Sath: 24 000 : 2400 = **10 cm**.
(15 — akvarium to'la deb hisoblangan.)

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9. Yarmi: sath 30 cm, suv 80 · 50 · 30 = 120 000 cm³ = 120 l.
Vaqt: 120 : 20 = **6 min**.
(12 min — idish to'liq to'ladigan vaqt.)

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10. Suv: 30 · 20 = 600 l = 600 000 cm³. Tubi: 120 · 50 = 6000 cm².
Balandligi: 600 000 : 6000 = **100 cm**.

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11. Tubi: 10 · 4 = 40 m². Sath 12 : 40 = 0,3 m ga tushdi.
1,5 − 0,3 = **1,2 m**.
(1,8 — suv qo'shilgandek hisoblangan.)

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12. Kub qirrasi 2 cm, yog'i 4 cm². 4 ta kubda 24 ta yoq.
4 ta joyda kublar yopishgan: 4 · 2 = 8 ta yoq ichkarida. Tashqarida 24 − 8 = 16 ta yoq.
16 · 4 = **64 cm²**.
(96 — ichkaridagi yoqlar ham sanalgan.)

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13. 30 : 4 = 7 (qoldiq 2), 20 : 4 = 5, 10 : 4 = 2 (qoldiq 2).
7 · 5 · 2 = **70**.
(93 — hajmlar bo'lingan: 6000 : 64, lekin qoldiq joylarga butun kubcha sig'maydi.)

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14. Birinchi g'isht hajmi: 25 · 8 · 5 = 1000 cm³.
Kvadrat yuzi: 1000 : 10 = 100 cm². Tomoni: **10 cm**.
(50 — 100 ning yarmi.)

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15. Kub qirrasi: 36 : 3 = 12 cm. Kuboid: 36 cm, 12 cm, 12 cm.
Hajmi: 36 · 12 · 12 = **5184 cm³**.
(1728 — bitta kub hajmi.)

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16. Kub qirrasi 5 cm. Hosil bo'lgan kuboid: 12 + 5 = 17 cm, 5 cm, 5 cm.
Hajmi: 17 · 5 · 5 = **425 cm³**.
(300 — faqat tosh blok.)

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17. Suv: 300 · 40 = 12 000 cm³. Ikki tub birga: 300 + 500 = 800 cm².
Sath: 12 000 : 800 = **15 cm**.
(20 — 40 ning yarmi, 24 — suv faqat B ga bo'lingan.)

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18. $\frac{1}{5}$ qismi: 36 : 3 = 12 l. To'la idish: 12 · 5 = 60 l.
Yana kerak: 60 − 36 = **24 l**.
(60 l — butun idish.)

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19. Tubi: 45 · 40 = 1800 cm². Sath 20 − 12 = 8 cm ko'tarildi.
1800 · 8 = 14 400 cm³ = **14,4 l**.
(36 l — idishdagi hamma suv.)

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20. Qirrasi: 32 · 32 · 32 = 32 768, demak 32 cm.
Yog'i: 32 · 32 = **1024 cm²**.
(32 — bu qirra.)
